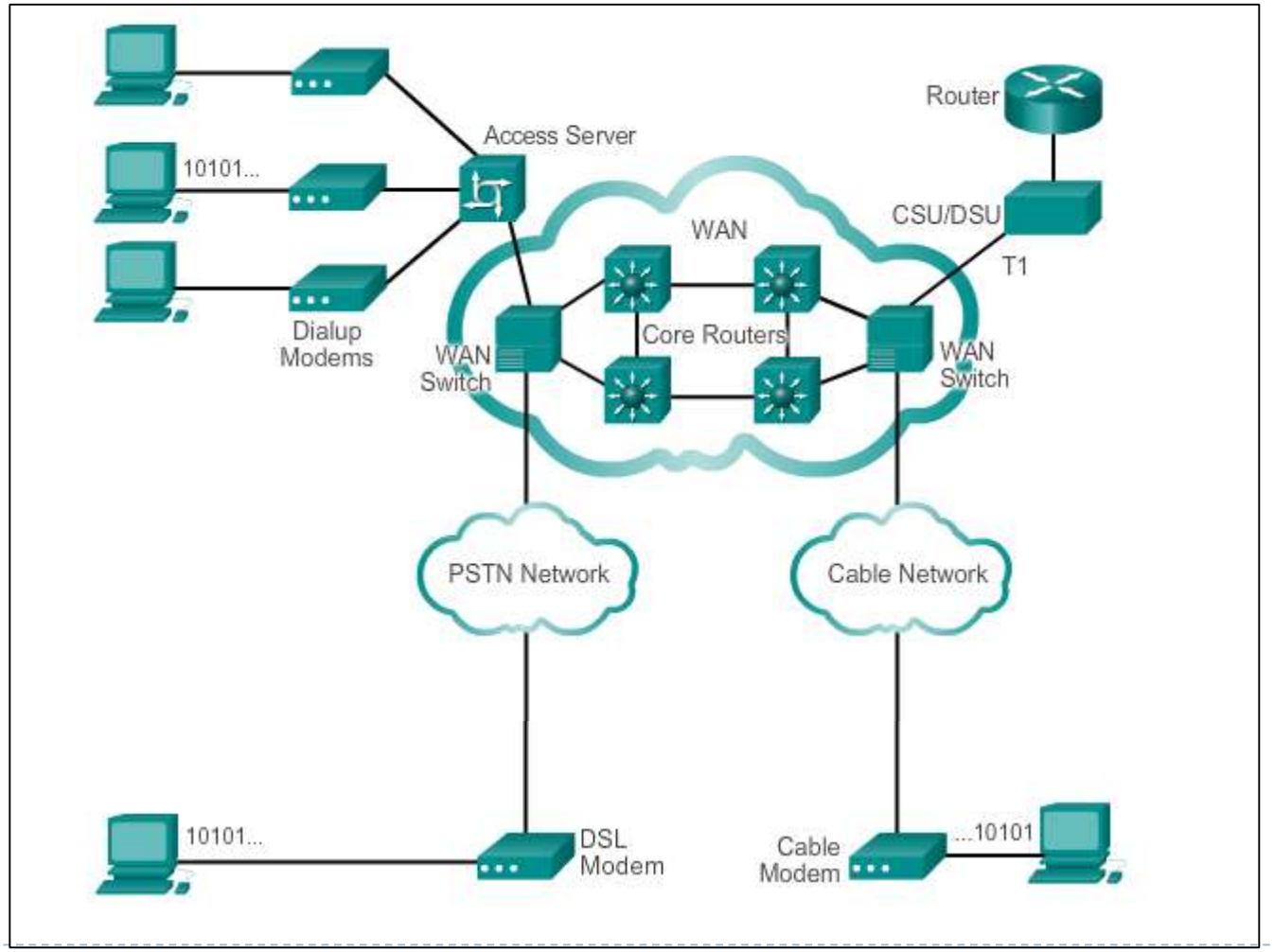
Lecture 4: WAN Devices &Technology

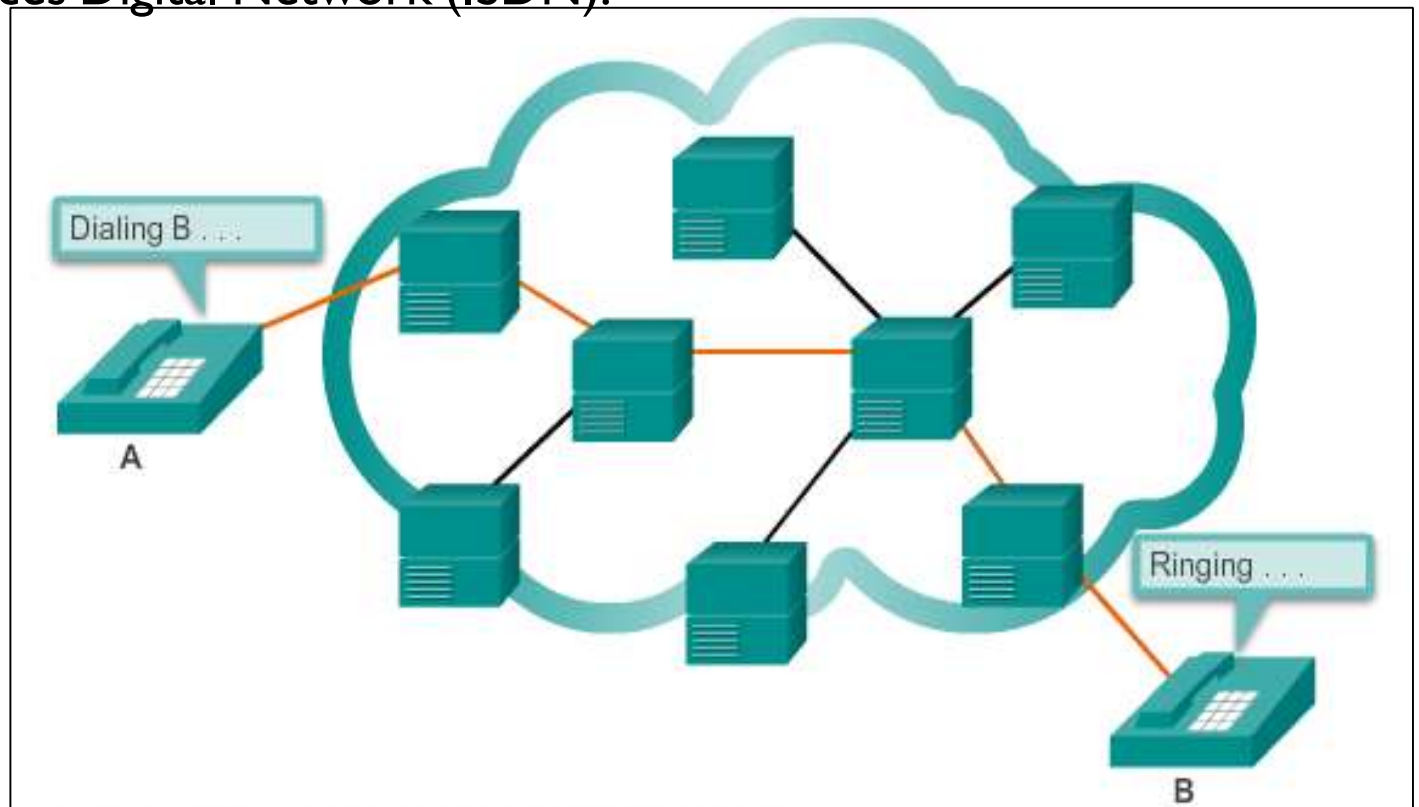
Selecting a WAN Technology

WAN Devices


Router
Access Server
10101...
CSU/DSU
WAN
T1
Dialup
Modems
Core Routers
WAN
Switch
WAN
Switch
PSTN Network
Cable Network
10101...
DSL
Modem
Cable
Modem
...10101

Circuit Switching

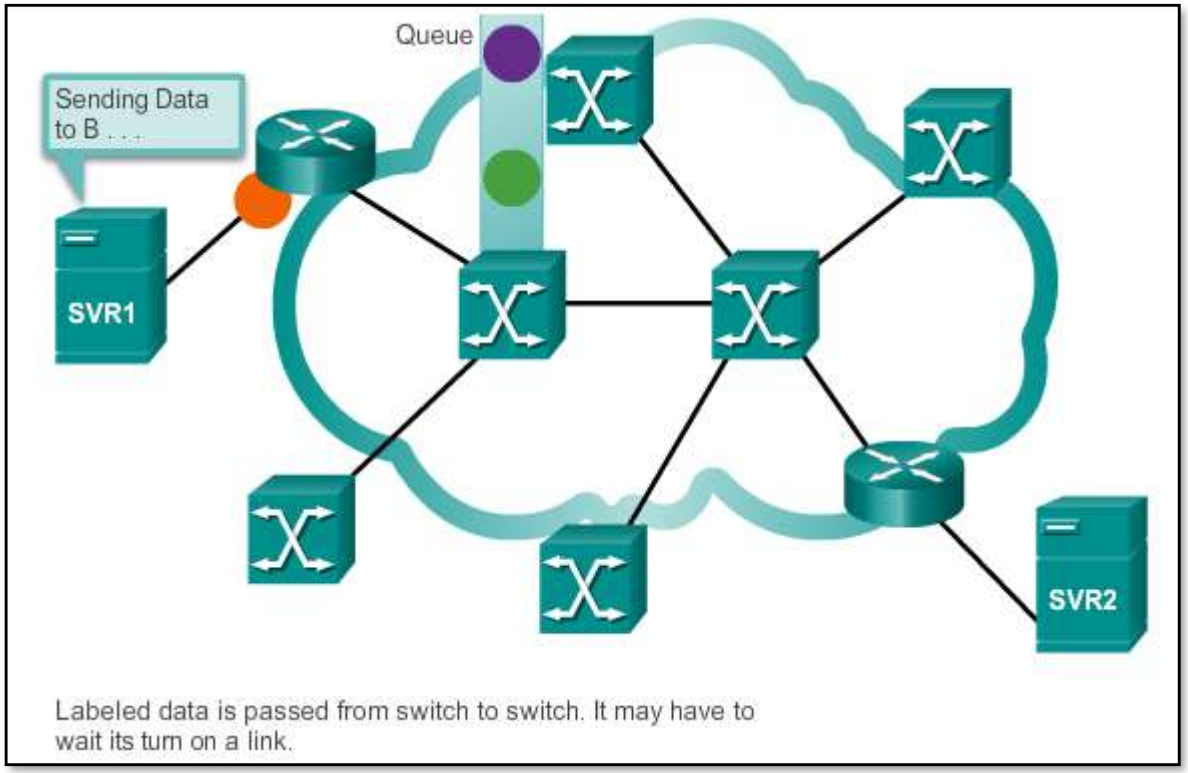
The two most common types of circuit-switched WAN technologies are the public switched telephone network (PSTN) and the Integrated Services Digital Network (ISDN).



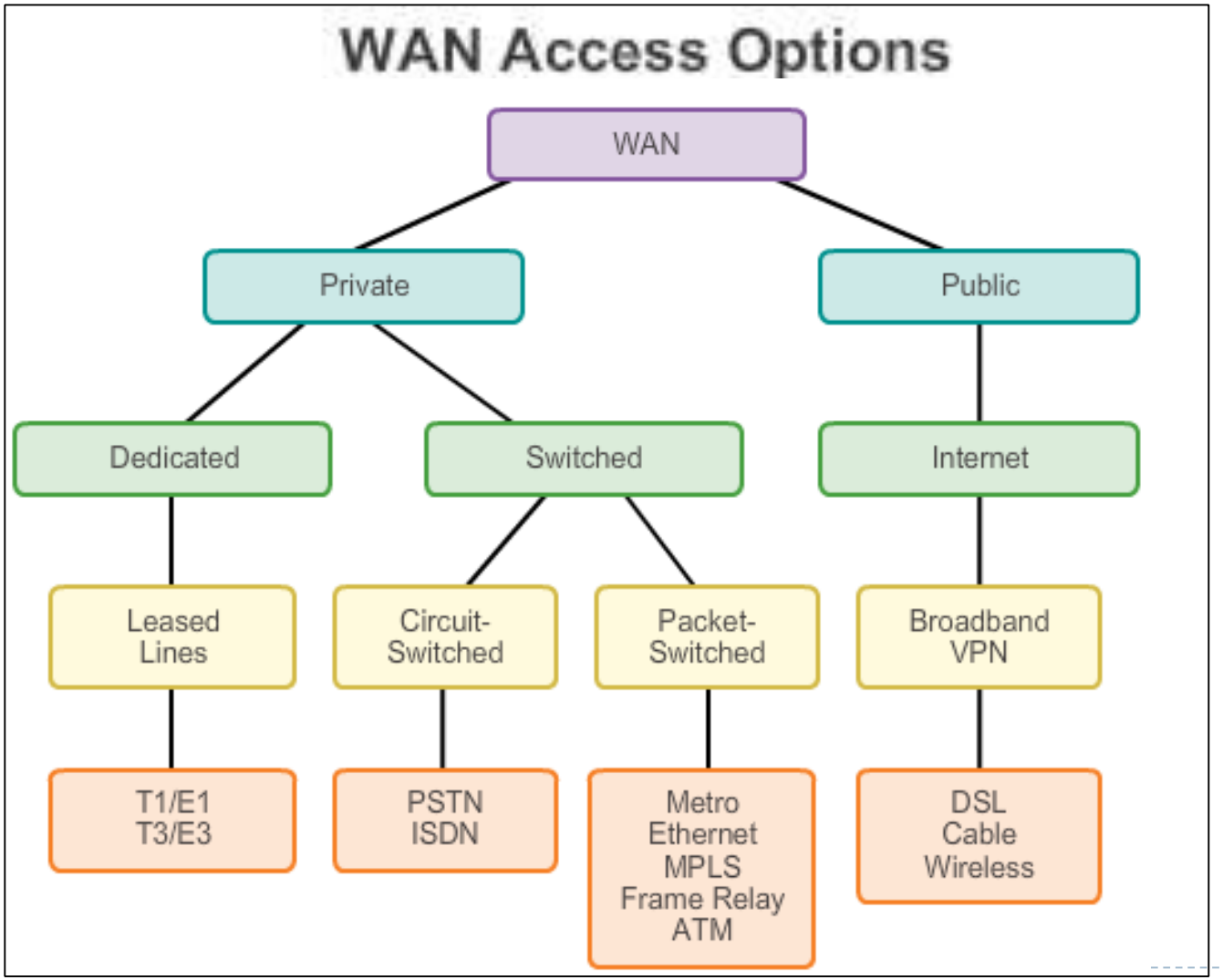
Dialing sets up a physical circuit through the system.

Packet Switching

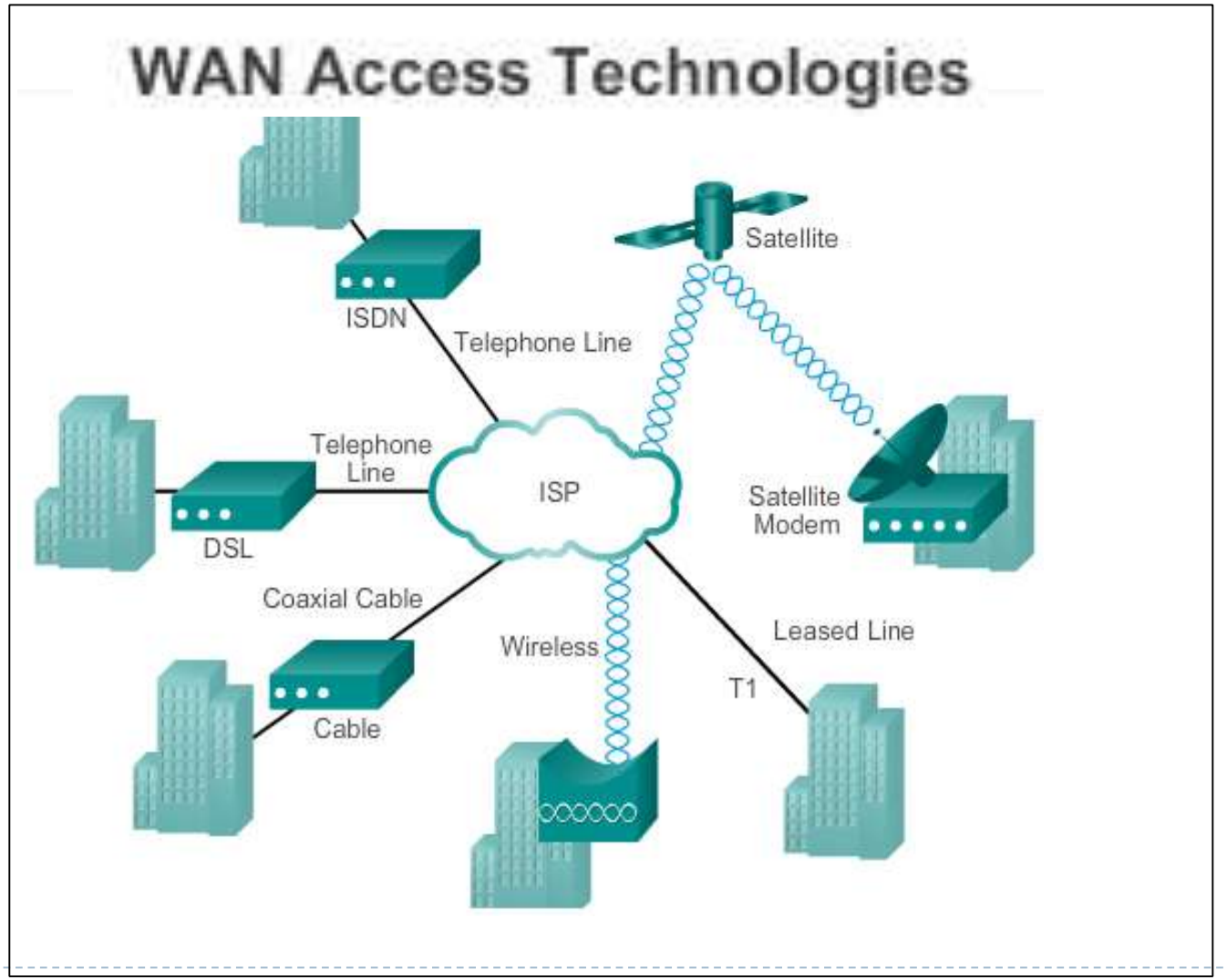
Splits traffic data into packets that are routed over a shared network. Packet-switching allow many pairs of nodes to communicate over the same channel.



WAN Link Connection Options



Service-Provided Network Infrastructure



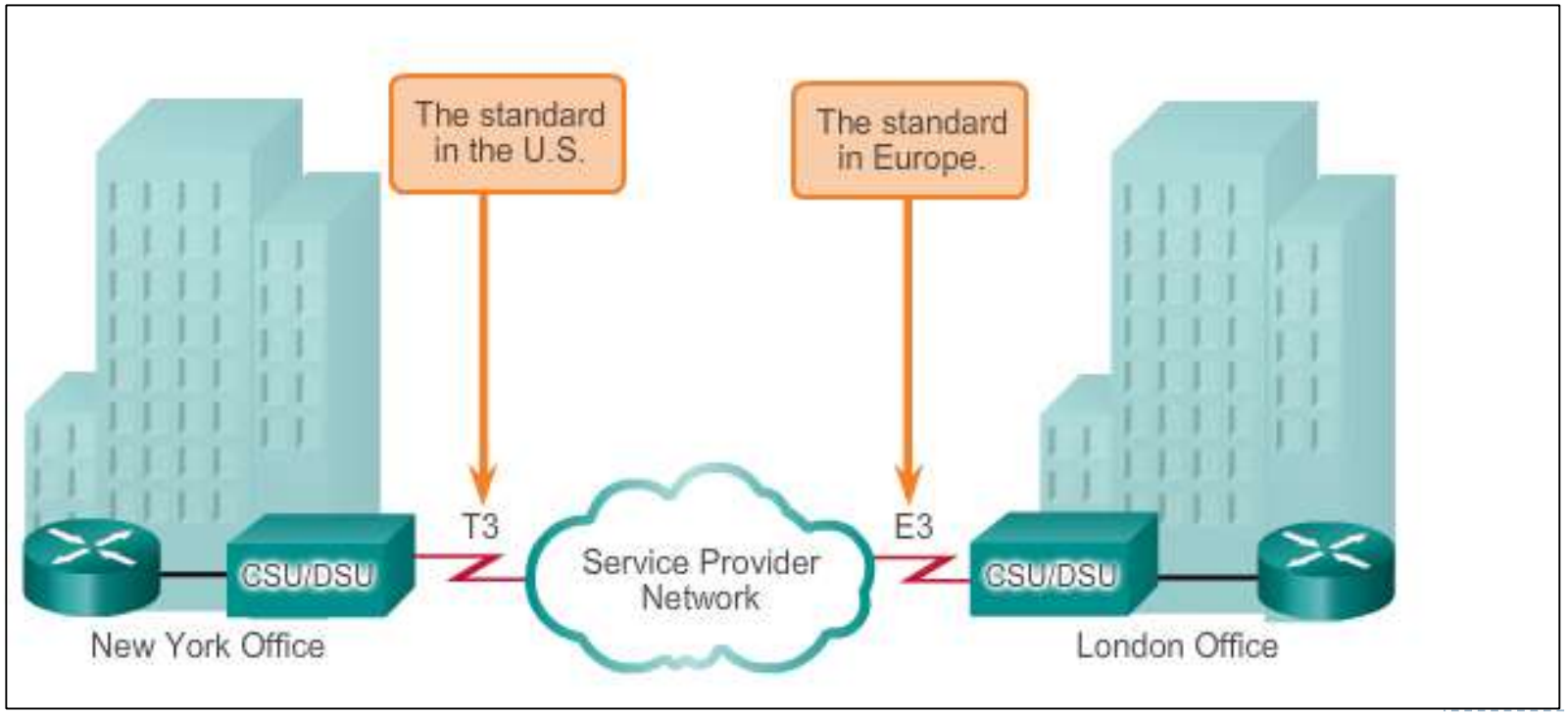
Leased Lines

Advantages:

- Simplicity
- Quality
- Availability

Disadvantages:

- Cost
- Limited flexibility



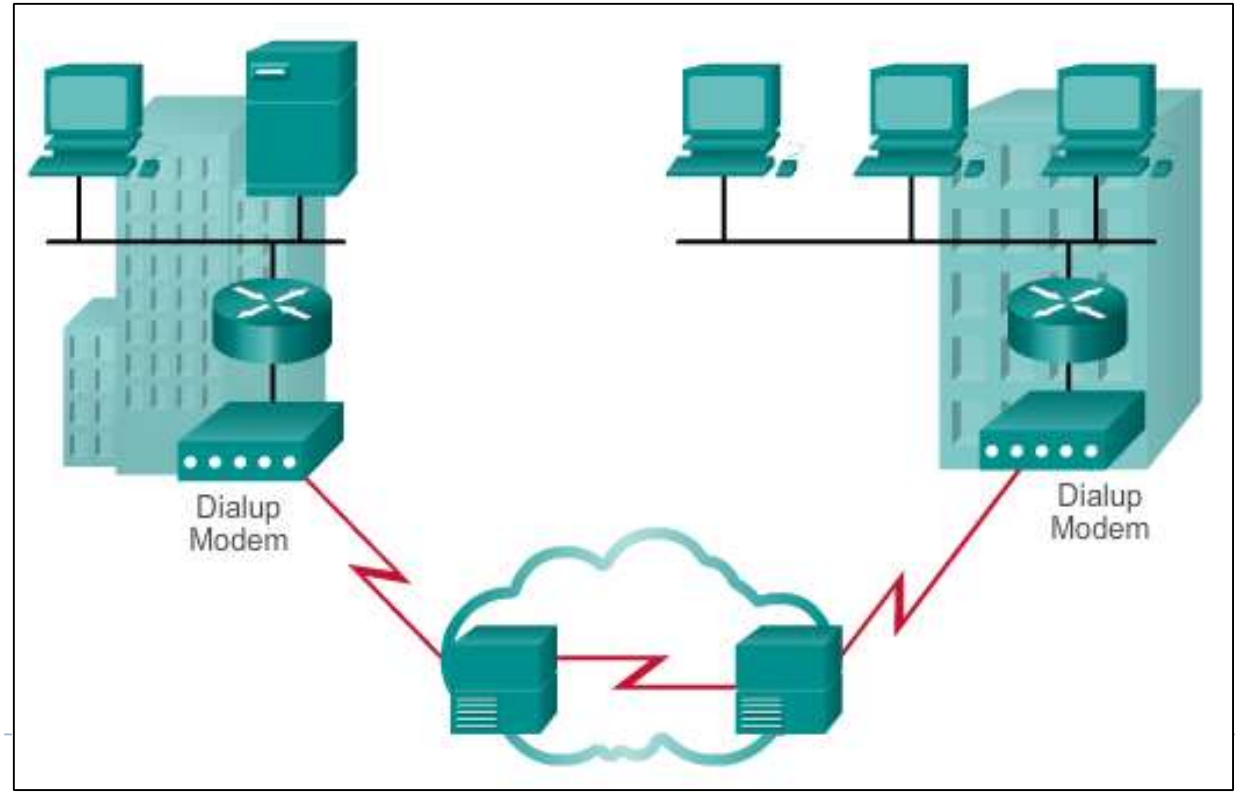
Dialup

Advantages:

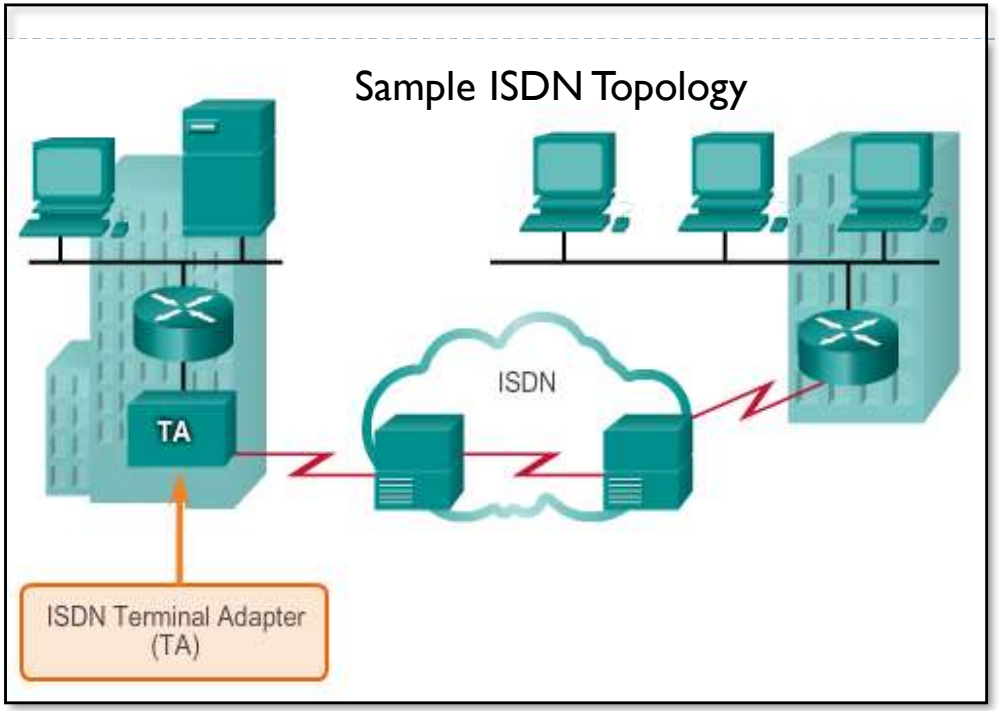
- Simplicity
- Availability
- Low implementation cost

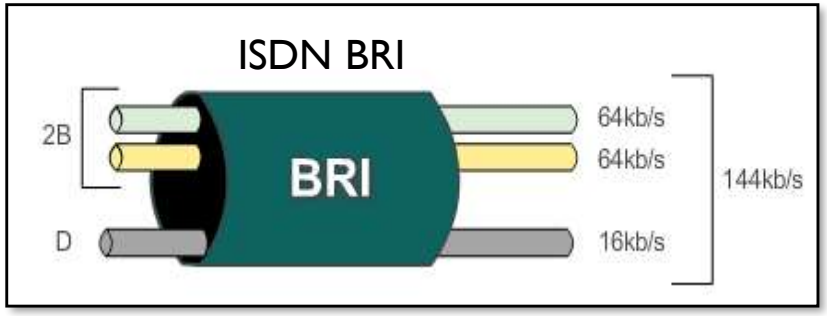
Disadvantages:

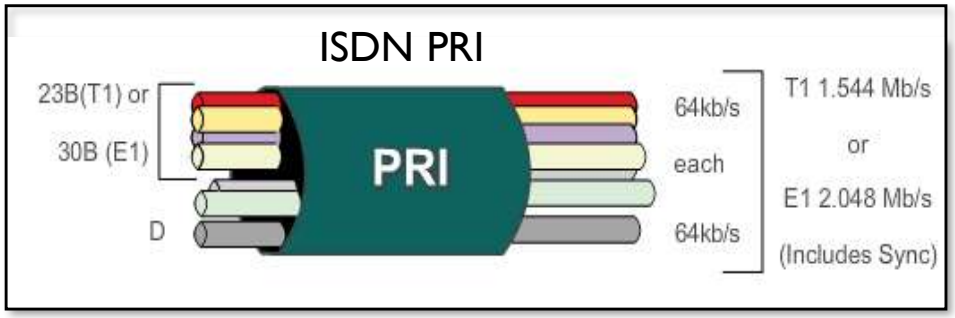
- Low data rates
- Relatively long connection time



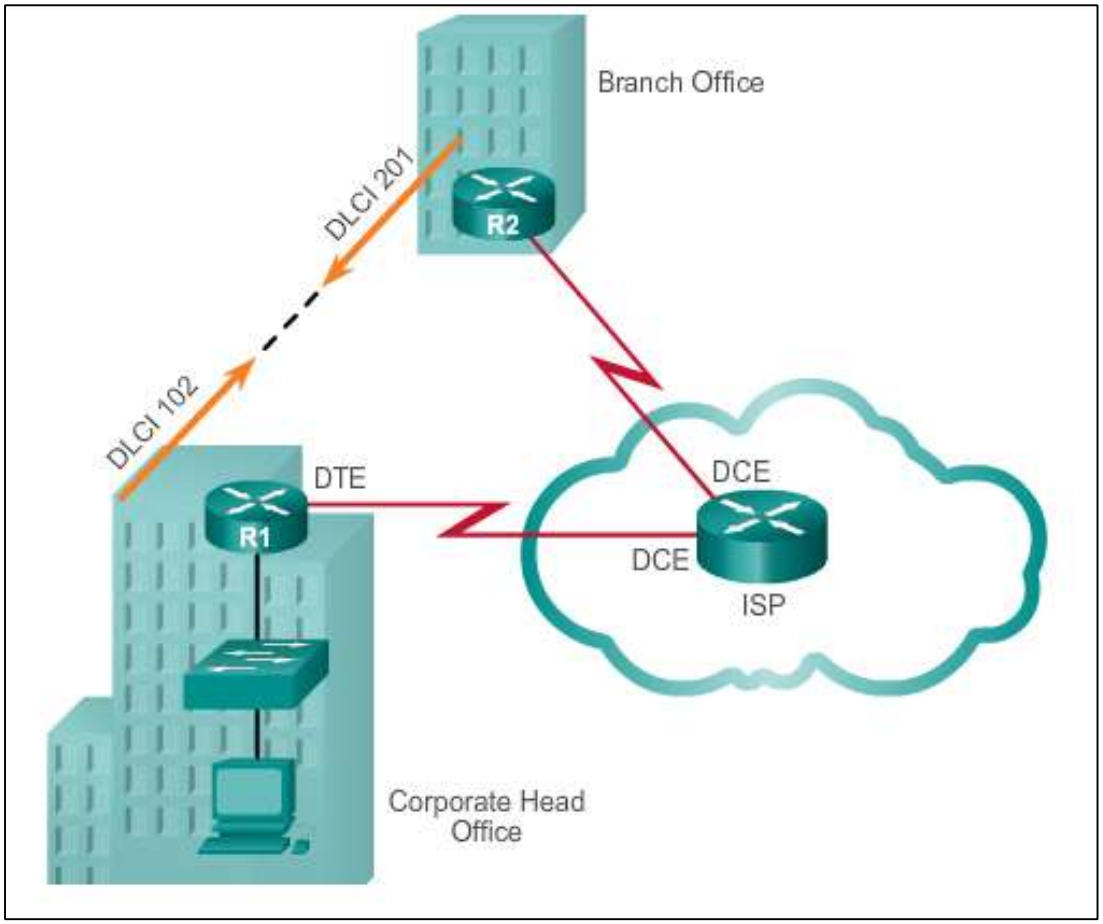
ISDN


Sample ISDN Topology
ISDN
TA
ISDN Terminal Adapter
(TA)


ISDN BRI
2B
D
BRI
64kb/s
64kb/s
16kb/s
144kb/s


ISDN PRI
23B(T1) or
30B (E1)
D
PRI
64kb/s
each
64kb/s
T1 1.544 Mb/s
or
E1 2.048 Mb/s
(Includes Sync)

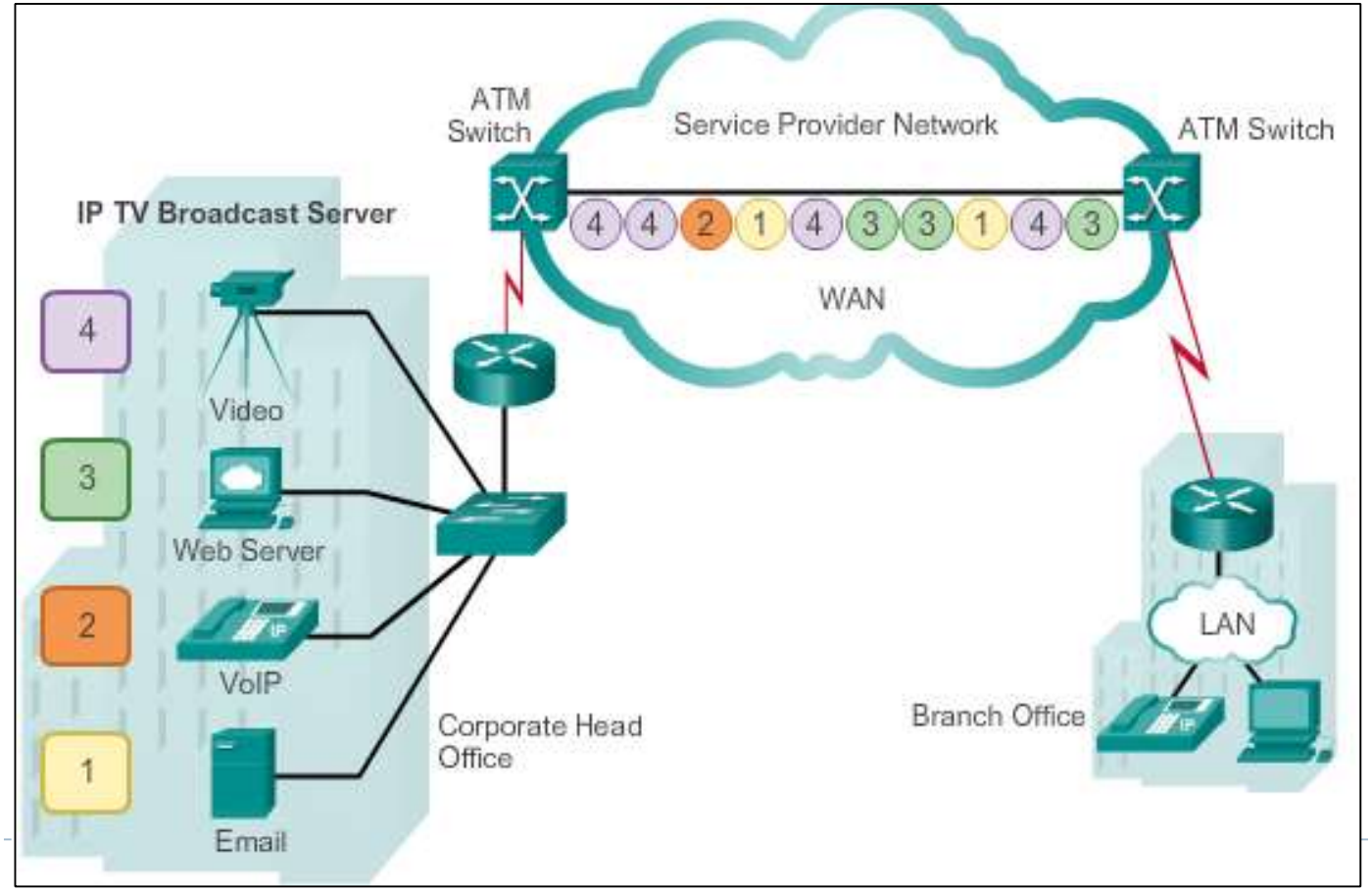
Frame Relay



- PVCs carry both voice and data traffic.
- PVCs are uniquely identified by a data-link connection identifier (DLCI).
- PVCs and DLCIs ensure bidirectional communication from one DTE device to another.
- R1 uses DLCI 102 to reach R2 while R2 uses DLCI 201 to reach R1.

ATM

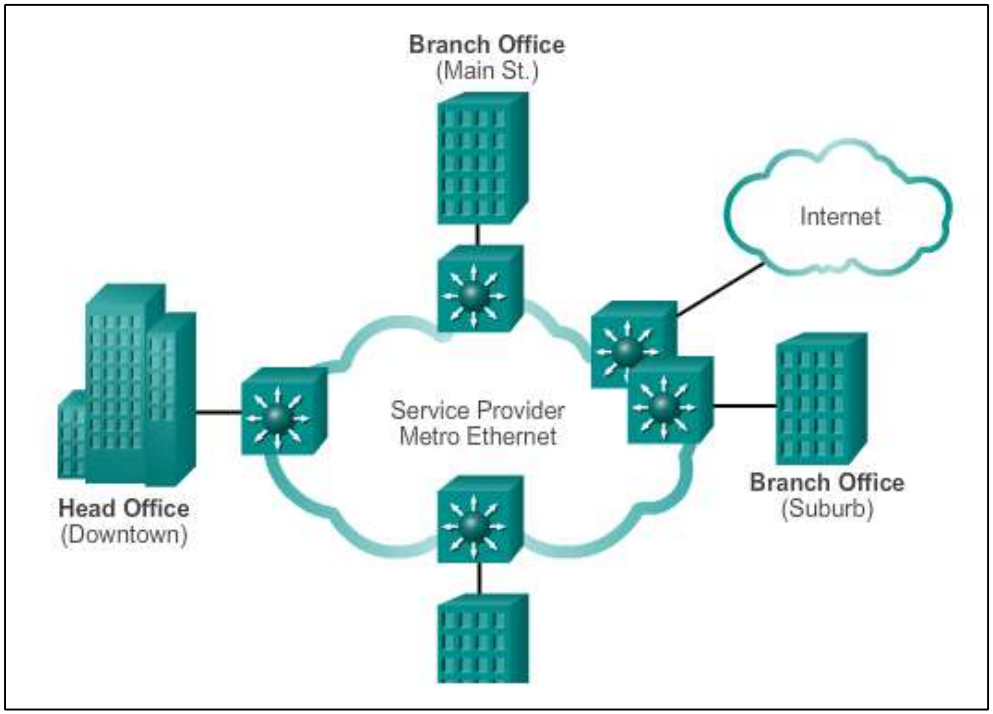
Built on a cell-based architecture, rather than on a frame-based architecture.ATM cells are always a fixed length of 53 bytes.



Ethernet WAN

Features and Benefits of Ethernet WAN include:

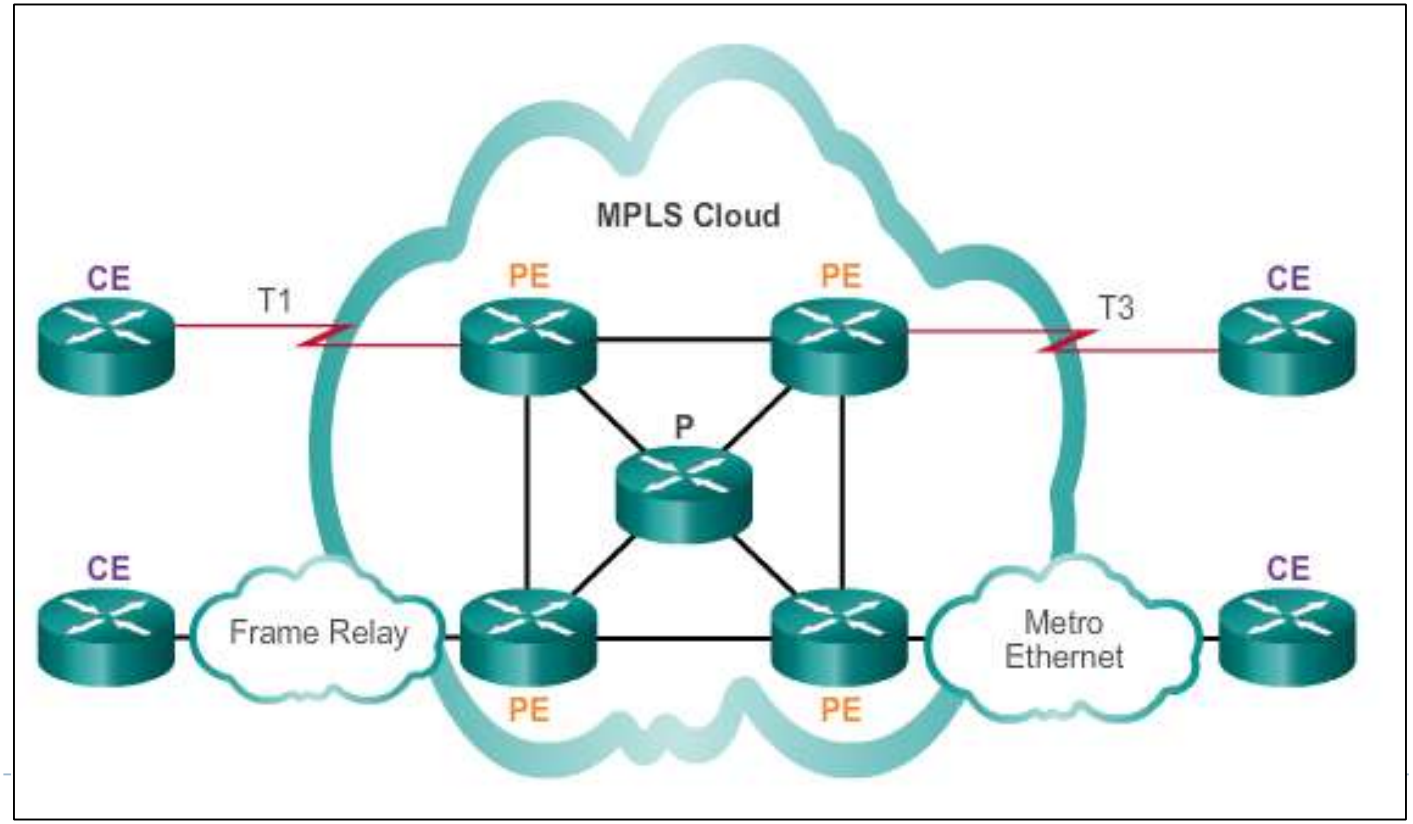
- Reduced expenses and administration
- Easy integration with existing networks
- Enhanced business productivity
- Service providers now offer Ethernet WAN service using fiber-optic cabling.
- Known as Metropolitan Ethernet (MetroE), Ethernet over MPLS (EoMPLS), and Virtual Private LAN Service (VPLS).



Note: Commonly used to replace the traditional Frame Relay and ATM WAN links.

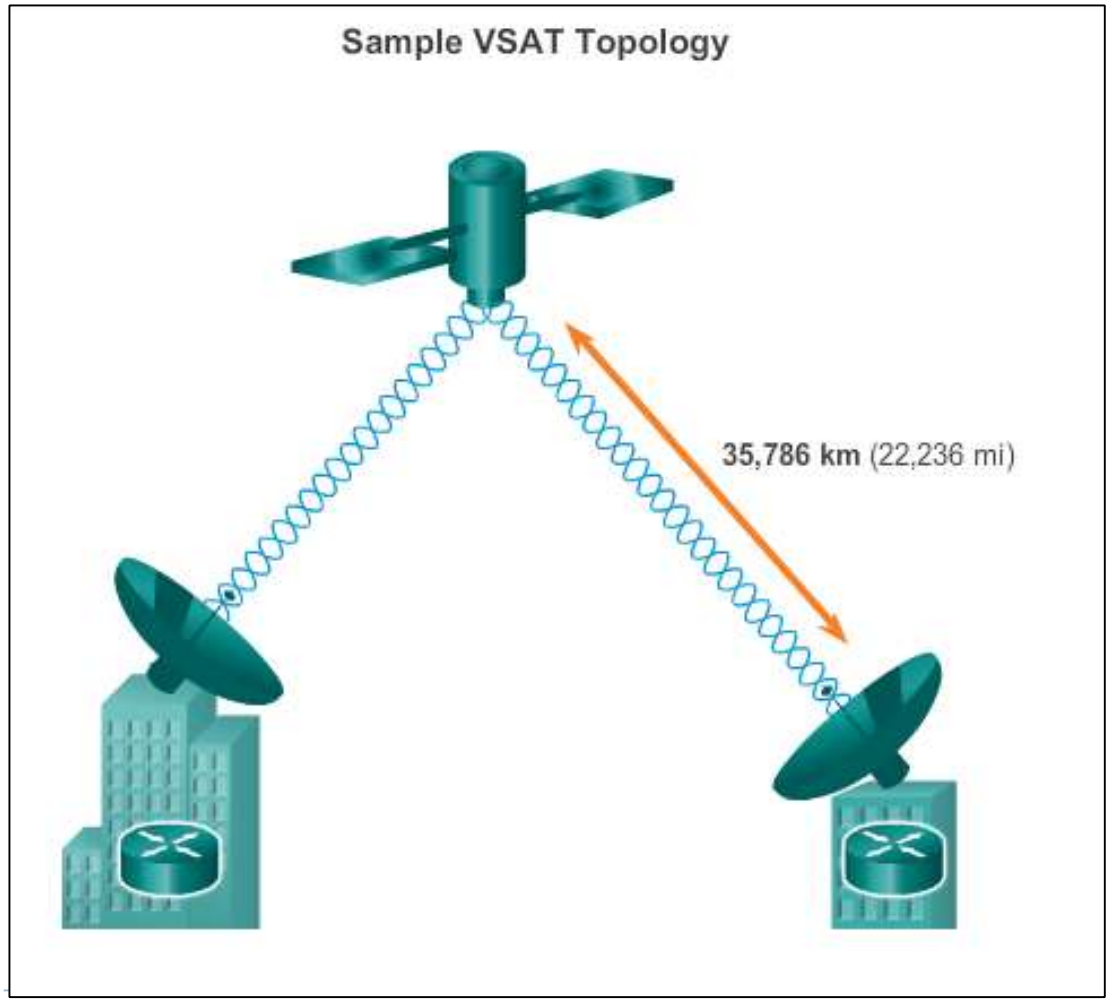
MPLS

Multiprotocol Label Switching (MPLS) is a multiprotocol high-performance WAN technology that directs data from one router to the next, based on short path labels rather than IP network addresses.



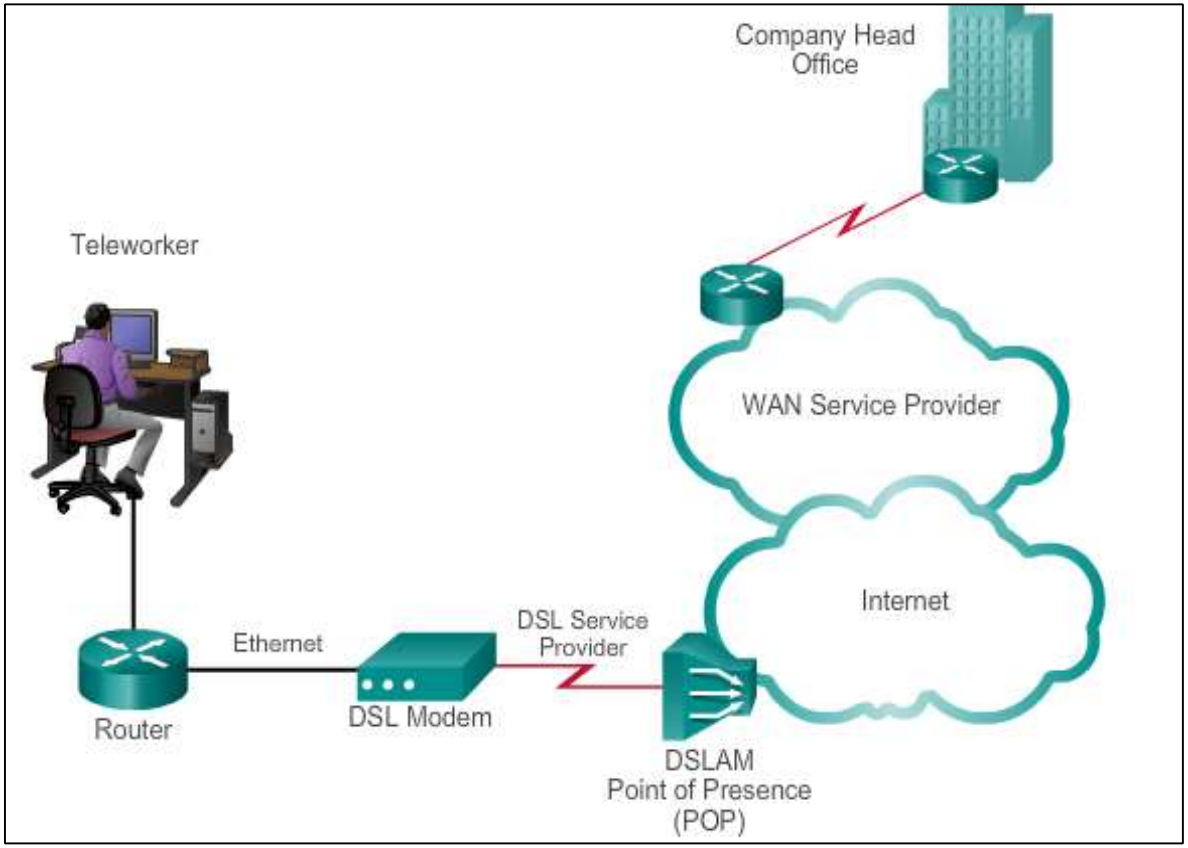
VSAT

Very small aperture terminal (VSAT) - a solution that creates a private WAN using satellite communications.



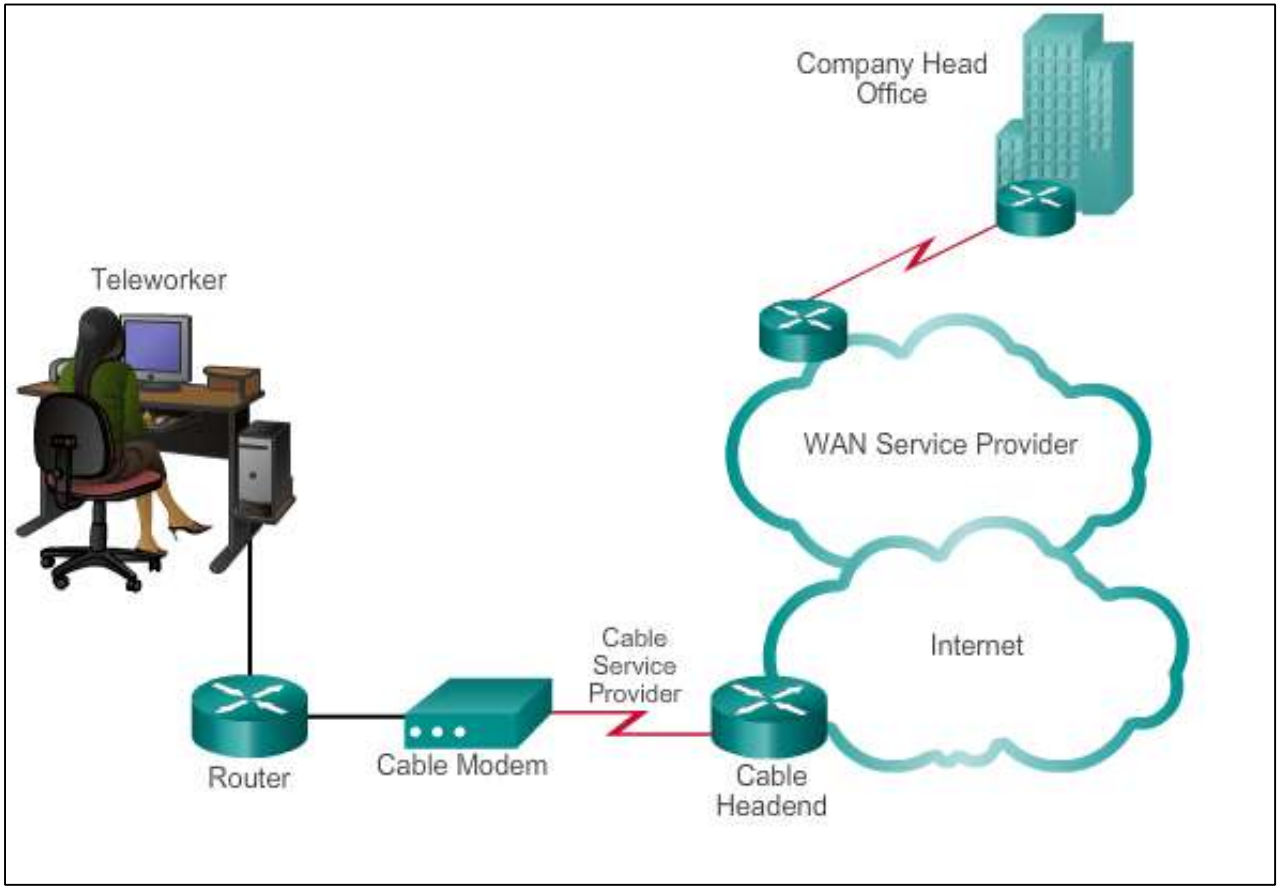
DSL

- Always-on connection technology that uses existing twisted-pair telephone lines to transport high-bandwidth data, and provides IP services to subscribers.
- A DSL modem converts an Ethernet signal from the user device to a DSL signal, which is transmitted to the central office.



Cable

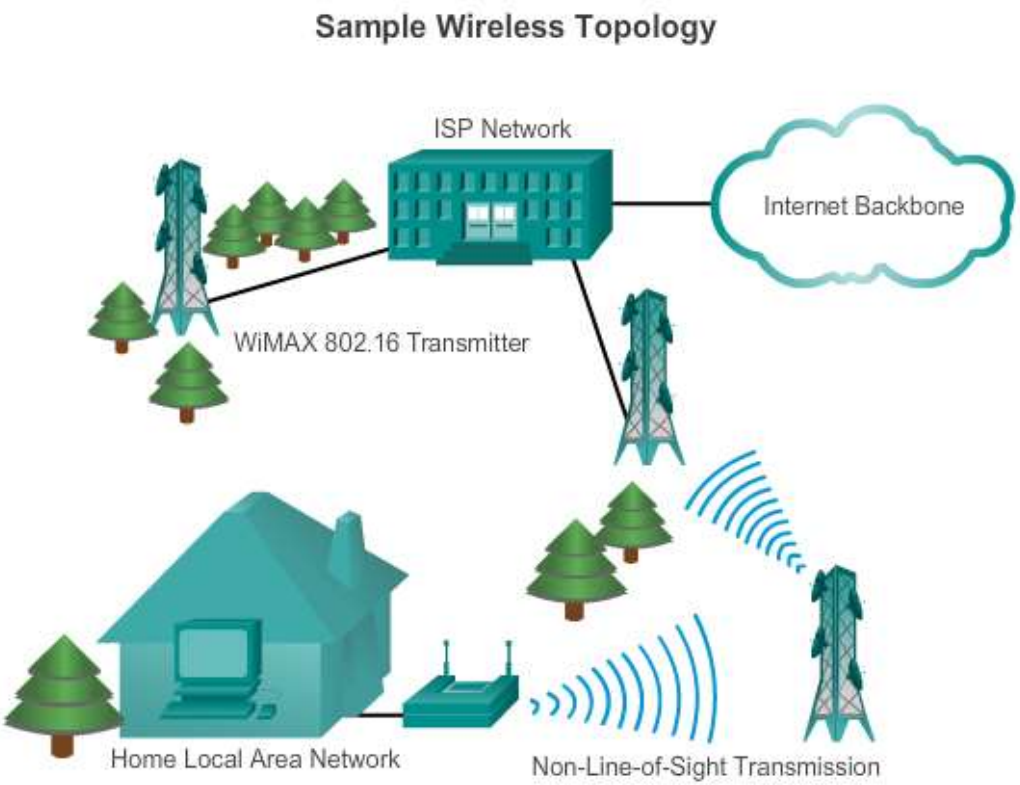
- Network access is available from some cable television networks.
- Cable modems provide an always-on connection and a simple installation.



Wireless

New developments in broadband wireless technology:

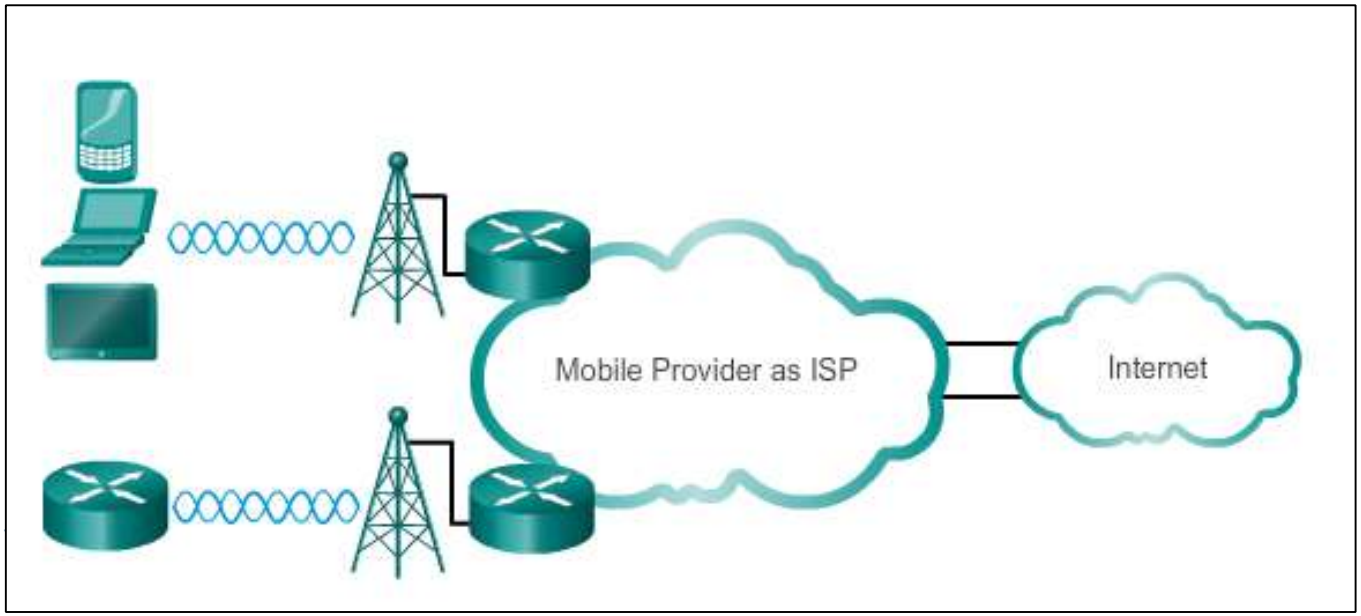
- **Municipal Wi-Fi** – Many cities have begun setting up municipal wireless
- **WiMAX** – Worldwide Interoperability for Microwave Access (WiMAX) is a new technology that is just beginning to come into use.
- **Satellite Internet**



3G/4G Cellular

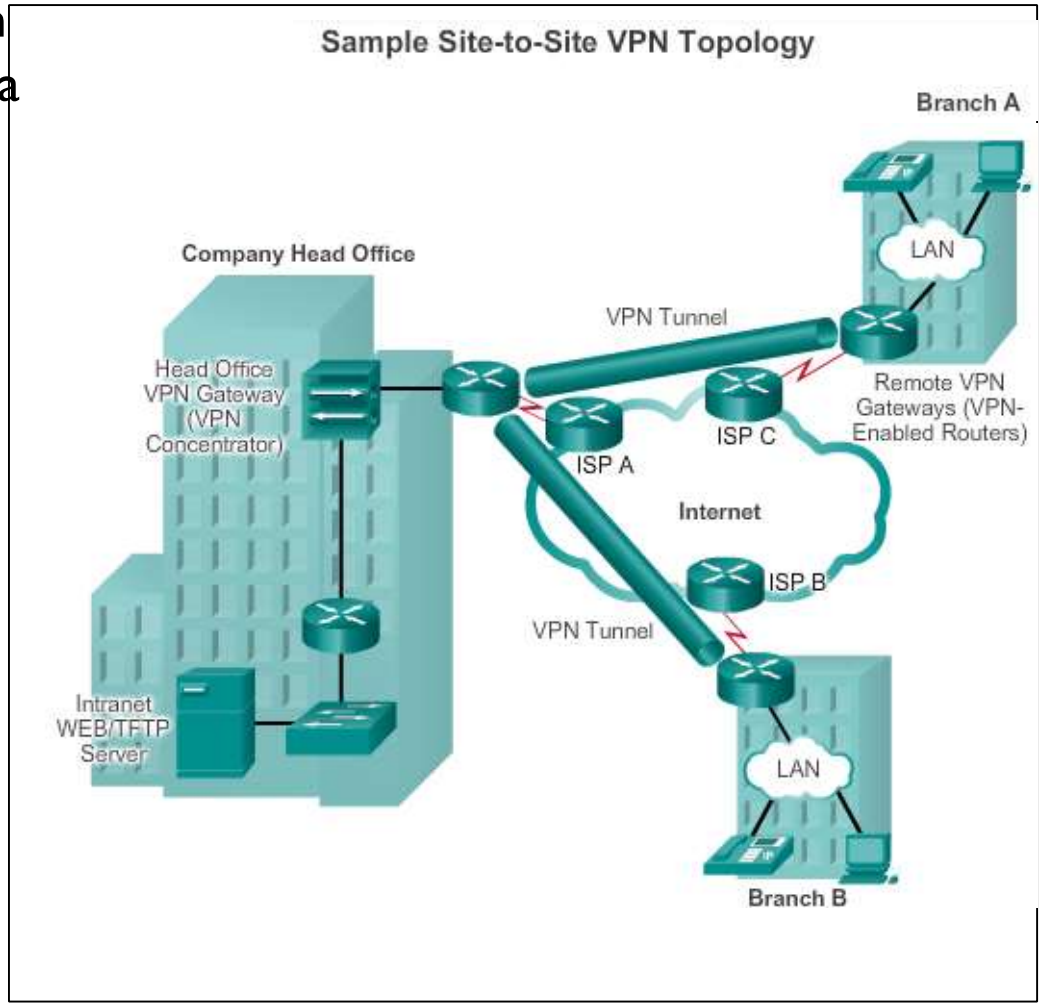
Common cellular industry terms include:

- **3G/4G Wireless** – Abbreviation for 3rd generation and 4th generation cellular access. These technologies support wireless Internet access.
- **Long-Term Evolution (LTE)** – A newer and faster technology, considered to be part of the 4th generation (4G) technology.



VPN Technology

VPN is an encrypted connection between private networks over a public network.



Benefits:

- Cost savings
- Security
- Scalability
- Compatibility with broadband technology

Two types of VPN:

- Site-to-site VPNs
- Remote-access VPNs

Choosing a WAN Link Connection

Answer the following questions when choosing a WAN Connection:

- What is the purpose of the WAN?
- What is the geographic scope?
- What are the traffic requirements?

Choosing a WAN Link Connection

- Should the WAN use a private or public infrastructure?
- For a private WAN, should it be dedicated or switched?
- For a public WAN, what type of VPN access is required?
- Which connection options are available locally?
- What is the cost of the available connection options?

Summary

- A business can use private lines or the public network infrastructure for WAN connections.
- WAN access standards operate at layers 1 and 2 of the OSI model, and are defined and managed by the TIA/EIA, ISO, and IEEE.
- A WAN may be circuit-switched or packet-switched.
- There is common terminology used to identify the physical components of WAN connections and who, the service provider or the customer, is responsible for which components.
- Service provider networks are complex and the service provider's backbone networks consist primarily of high-bandwidth fiber optic media.

Summary (cont.)

- Permanent, dedicated point-to-point connections are provided by using leased lines.
- Public infrastructure connections include DSL, cable, wireless, and 3G/4G cellular.
- Security over public infrastructure connections can be provided by using remote-access or site-to-site VPNs.